

Starting state

precondition.c state machine

Global behavior:

- Track activation button presses, send `EV_TOGGLE`
- Track power status frame, send `EV_CAR_(READY|NOT_READY)` on edges
- Track precondition status frame, send `EV_STATUS_(IDLE|STARTED|STARTING)` on every status frame
- Track battery temperature for web UI

enter:

If in persistent mode, and the toggle restored from flash is on, go to **MANAGED**

IDLE

event handler:
On `EV_TOGGLE`, go to **REQUESTED**

REQUESTED

CAR_START_DELAY
tick:
Wait 8 seconds, then re-enter **REQUESTED** to begin sending start burst

START_BURST
event handler:
Track `EV_STATUS` events
tick:
Send start message for 6 ticks, then:
If only seen **IDLE**:
- `WAIT_STARTING`
If seen **STARTING**:
- `WAIT_STARTED`
If seen **STARTED**:
- `ACTIVE/MANAGED`

WAIT_STARTING
event handler:
`EV_STATUS_STARTING`:
- `WAIT_STARTED`
`EV_STATUS_STARTED`:
- `ACTIVE/MANAGED`
tick:
10 seconds (since **START_BURST**) without seeing `EV_STATUS_(STARTING|STARTED)`:
If we need to retry:
- `START_BURST`
If we're out of retries, in Once mode:
- **STOPPING**
If we're out of retries, not in Once mode:
- **MANAGED**

ACTIVE
event handler:
`EV_STATUS_IDLE`:
- **STOPPING**
`EV_STATUS_STARTING`:
- `WAIT_STARTED`

WAIT_STARTED
event handler:
`EV_STATUS_STARTED`:
- `ACTIVE/MANAGED`
tick:
70 seconds (since **START_BURST**) without seeing `EV_STATUS_STARTED`:
If we need to retry:
- `START_BURST`
If we're out of retries, in Once mode:
- **STOPPING**
If we're out of retries, not in Once mode:
- **MANAGED**

MANAGED
event handler:
Track `EV_STATUS` events;
`EV_CAR_READY`:
In persistent mode, `CAR_START_DELAY`
tick:
If last seen status was **IDLE** + car is ready, wait 5 minutes since last **IDLE** rising edge, then re-enter **REQUESTED**

STOPPING

STOP_BURST
tick:
Send stop message for 6 ticks, then go to **WAIT_STOPPED**

WAIT_STOPPED
event handler:
On `EV_STATUS_IDLE`, go to **IDLE**
tick:
While retries are left, go to **STOP_BURST** after 10 seconds since last retry.
If retries are exhausted, go to **IDLE** unconditionally

